

SimpleSFT Project Report

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Abstract

SimpleSFT estimates whether supervised fine-tuning jobs for popular research LLMs, including Qwen-family models, will fit on available GPU resources before launch. The project combines a typed analytical memory model, CUDA-based measurement tools, saved benchmark comparisons, and configuration search. A user-friendly web UI recommends training strategies based on the estimator's output and exports a TRL-compatible YAML strategy for the selected configuration. The current estimator shows a global mean absolute error of 1.332 GiB and a median relative error of 6.1 percent on a saved measurement corpus, which is strong for pre-launch screening when users keep 10% headroom.

1 Project Goal

SimpleSFT estimates the memory requirements for supervised fine-tuning (SFT) jobs on popular large language models (LLMs) used in academic research. These models include open-weight Qwen-family models, OLMo, LLaMA, and Mistral, which are common in current research workflows and span sizes where a small change in distributed strategy can determine whether a job runs at all.

Given a model, context length, resource pool, and training strategy, estimate the expected per-rank peak memory and identify the training phase that produces that peak.

Today, users face slow trial-and-error development workflows to find parallel strategies that meet their specific needs: short debug allocations on OSC allow only one GPU, while larger allocations can require long queues or long reservations. SimpleSFT helps users choose a plausible strategy before occupying GPUs for extended setup runs.

The state of OSC queues is often unpredictable and frustrating. This has led to a culture of overprovisioning, where users request more resources than they expect to need just to be sure they will have time to debug their jobs. This is inefficient for everyone: users wait longer in queues, and the cluster is less available for other work. SimpleSFT provides a more informed way to request resources, which can reduce wait times and improve overall cluster efficiency.

This framing matters on shared academic clusters. An NLP researcher seeking to do SFT often begins with a concrete scientific question, a model family, and some training data. The act of engineering a system to run SFT efficiently is incidental to the true research question. SimpleSFT lightens that work by providing a framework for getting to model training with less trial and error.

2 System Overview

The project has four main pieces:

- **Model inspection:** Hugging Face models are summarized as a `ModelSpec`: layer count, hidden and intermediate sizes, vocabulary, attention layout, parameter shapes, trainable linear layers, tensor-layout roles, and optional architecture metadata.
- **Analytical estimation:** an `EstimatorConfig` describes the training strategy. The estimator builds resident state, retained activations, workspace windows, and phase peaks, then returns a `MemoryResult`.
- **Measurement and saved comparisons:** CUDA microsteps provide ground-truth memory measurements for selected configurations. The saved measurements let later estimator changes be checked against the same cases.
- **Workflow tools:** CLI and local web paths expose estimate, search, measure, compare, benchmark, report, YAML export, and SFT launch commands.

3 Estimator Model

SimpleSFT treats a training step as three candidate peaks:

$$M_{\text{peak}} = \max(M_{\text{forward}}, M_{\text{backward}}, M_{\text{optimizer}}).$$

Forward starts from parameters and runtime support. Backward adds gradients and backward-live activation state. The optimizer step adds update scratch, communication buckets, and phase carryover. This breakdown gives a useful structure for model design and calibration because each estimator change can be attached to a specific phase and memory family.

3.1 Inputs

The estimator combines a `ModelSpec` and an `EstimatorConfig`. The model spec supplies architecture and tensor metadata: parameter shapes, linear-layer roles, attention widths, vocabulary size, layer count, and tensor-parallel layout annotations. The config supplies the training strategy: tuning mode, optimizer, dtypes, sequence length, GPU budget, attention backend, checkpointing, LoRA settings, distributed mode, tensor and sequence parallel choices, runtime-support assumptions, and DDP/ZeRO bucket sizes.

The inspected model metadata is highly detailed. For a Qwen-style model, the estimator needs the number of transformer blocks, the hidden width, the MLP expansion width, the number of attention heads, the number of key/value heads, the vocabulary projection shape, and the local role of each large matrix under tensor parallelism. The same total parameter count can lead to different memory behavior when the attention projection layout, MLP layout, or vocabulary tensor is sharded differently.

Estimator input	Why it affects peak memory
Parameter shapes	Drive parameter bytes, gradient bytes, optimizer state, and tensor-parallel local ownership.
Linear-layer list	Identifies LoRA target modules and adapter parameter counts.
Attention layout	Determines query and key/value widths, attention workspace, and long-context retained state.
Distributed strategy	Changes rank ownership of gradients, optimizer state, parameters, buckets, and gather windows.
Checkpointing flag	Changes retained activation state and adds backward recompute workspace.

3.2 Resident State

Resident state is the memory floor on one rank:

$$R_{\text{base}} = M_{\text{params}} + M_{\text{opt}} + M_{\text{master}} + M_{\text{runtime}} + M_{\text{persist}}, \quad R_{\text{grad}} = R_{\text{base}} + M_{\text{grads}}.$$

Parameters are counted from explicit parameter shapes when available, with tensor-parallel sharding applied to row, column, and vocabulary-parallel tensors. ZeRO-3 shards trainable parameters across data-parallel ranks. LoRA keeps the dense base weights resident and makes the adapter tensors trainable, so gradients and optimizer state scale with adapter size.

Optimizer memory is optimizer-specific. Adam and AdamW keep two state tensors, RMSProp and Adagrad keep one, SGD keeps none, and Adafactor uses factored state for matrix-like tensors. ZeRO-2 and ZeRO-3 shard optimizer state and gradients for tested optimizer paths. Runtime support is represented with named CUDA, allocator, NCCL, DeepSpeed, and backend-buffer terms.

The local tensor count is computed before bytes are assigned. Tensor-parallel metadata determines whether a parameter is replicated or divided across the tensor-parallel group. ZeRO then applies data-parallel sharding to the training state owned by its stage. This order preserves the difference between a matrix partitioned by tensor parallelism and a state object partitioned by ZeRO.

Object	Full fine-tuning	LoRA fine-tuning
Base weights	resident	resident
Trainable weights	dense model tensors	adapter tensors
Gradients	dense trainable tensors	adapter gradients
Optimizer state	dense trainable tensors	adapter optimizer state

This distinction is central for LLMs, because a full fine-tune must carry optimizer and gradient state for the dense model. A LoRA run still stores the base weights during forward and backward passes, while trainable-state memory demand is determined by the targeted projection layers and the selected adapter rank.

For one dense matrix $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, full fine-tuning makes all $d_{\text{out}}d_{\text{in}}$ elements trainable. LoRA inserts

$$\Delta W = \frac{\alpha}{r} BA, \quad A \in \mathbb{R}^{r \times d_{\text{in}}}, \quad B \in \mathbb{R}^{d_{\text{out}} \times r},$$

so the trainable count for that target becomes $r(d_{\text{in}} + d_{\text{out}})$. The dense base matrix remains part of resident parameter memory, while gradient and optimizer memory follow the adapter tensors. This gives the estimator a direct way to reason about rank, target module selection, and optimizer state without introducing a separate LoRA-only formula for the rest of the model.

Optimizer state is handled at parameter-tensor granularity. Adam-style optimizers attach two state tensors to each trainable tensor. Adafactor uses row and column factors for matrix-like tensors, which is materially smaller for large projection matrices. This tensor-level treatment matters for mixed training modes: a run can have dense frozen base parameters, trainable adapter matrices, and optimizer state only for those adapters.

3.3 Retained Activations and Attention

Retained activation memory is a sum of terms:

$$M_{\text{act}} = M_{\text{visible}} + M_{\text{saved inputs}} + M_{\text{mlp}} + M_{\text{attn}} + M_{\text{lora}} + M_{\text{ckpt}} + M_{\text{loss}} + M_{\text{widened}}.$$

Full fine-tuning grows with layer count, local token count, hidden width, MLP width, and attention-path width. LoRA reduces trainable activation pressure by moving gradients onto targeted adapter paths and low-rank intermediates. Gradient checkpointing stores fewer layer internals and shifts work into backward recomputation.

Attention backend matters most at long context length. Standard attention requires attention-score states that scale quadratically with sequence length. FlashAttention removes the need to store quadratic-length attention scores by using tiled computation and recomputation, with a smaller additional workspace term [1]. SimpleSFT therefore models FlashAttention primarily through retained attention-path context and workspace terms.

Gradient checkpointing is modeled as a change to the activation life cycle. Layer internals that would be retained until backward are reduced, while a separate recompute window is added during backward. The estimator represents this through checkpoint-boundary bytes, saved linear-input bytes, residual and normalization context, and recompute workspace. This keeps the memory tradeoff visible: checkpointing lowers retained forward state and raises backward compute plus transient workspace.

3.4 Workspace and Phase Assembly

Workspace is modeled as a live window. The main terms are attention workspace, backward kernel workspace, recompute workspace, loss workspace, tensor/sequence parallel communication windows, optimizer update scratch, and distributed bucket objects.

DDP contributes a reducer bucket and a communication-overlap window. ZeRO contributes all-gather, reduce, and prefetch buckets. Bucket capacities are derived from configured element counts and clipped by the local payload that can fill them. ZeRO optimizer pressure is the maximum of fetch, update, and communication subphases.

The bucket model uses configured capacities and local payload caps. A reducer or all-gather bucket can reserve a large object, yet the active payload remains bounded by the data held on that rank. This makes the estimate responsive to both runtime settings and model size. ZeRO-2 keeps parameters resident while sharding optimizer state and gradients. ZeRO-3 also shards trainable parameters, adding parameter gather windows around computation.

The final phase equations are:

$$\begin{aligned} M_{\text{forward}} &= R_{\text{base}} + A_{\text{fwd}} + W_{\text{fwd}}, \\ M_{\text{backward}} &= R_{\text{grad}} + A_{\text{bwd}} + W_{\text{bwd}}, \\ M_{\text{optimizer}} &= R_{\text{grad}} + E_{\text{bwd}} + W_{\text{opt}} + C_{\text{opt}}. \end{aligned}$$

Here E_{bwd} is backward-end state such as gradients and live distributed buckets, and C_{opt} represents carryover that remains live at optimizer-step entry.

The phase assembly makes the peak source explicit. A forward peak usually signals parameter plus retained activation pressure. A backward peak usually signals the combination of gradients, retained or recomputed activations, and communication workspace. An optimizer-step peak usually signals optimizer state, update scratch, and ZeRO communication windows. Reporting this phase is important because two configurations can have similar total peaks while needing different fixes. For example, checkpointing helps a forward-activation peak, while ZeRO stage changes help persistent optimizer or gradient pressure.

Peak phase	Dominant terms SimpleSFT reports
Forward	Base resident state, retained activations, attention workspace, loss-related state.
Backward	Resident state with gradients, backward activation state, checkpoint re-compute workspace, DDP or ZeRO communication windows.
Optimizer step	Gradients, optimizer state, update scratch, ZeRO fetch and communication windows, carried state from backward.

4 Strategy Selection and User Workflow

The estimator can answer one fixed-configuration question or search over a small set of feasible strategies. A candidate is an `EstimatorConfig` paired with a resolved `ModelSpec`. The CLI search path evaluates the requested resource and distributed-mode options, separates feasible and infeasible cases, and ranks candidates by feasibility, distributed-mode simplicity, and memory headroom.

The simplicity order is:

$$\text{tp_only} \prec \text{ddp} \prec \text{zero2} \prec \text{zero3}.$$

This favors a smaller distributed stack when the memory estimate permits it. The UI uses a richer candidate set over distributed mode, checkpointing, tensor-parallel degree, and sequence-parallel settings. DDP and ZeRO require enough GPUs for data parallelism after tensor parallelism is assigned, while sequence parallelism requires tensor parallelism.

Candidate generation follows the same constraints a user would apply by hand. The resource pool determines the available GPU count and memory budget. Tensor parallelism consumes GPUs inside each model replica. Data parallelism then uses the remaining replica count. ZeRO stages are considered only when a data parallel group exists. Sequence parallelism is paired with tensor parallelism, since it partitions sequence-local activation state across the tensor-parallel group.

Strategy family	Search treatment
Single GPU	Used when only one GPU is specified.
Tensor parallel	Shards eligible matrices and associated tensor-local state across the tensor-parallel group.
DDP	Replicates model state and changes communication workspace through reducer buckets.
ZeRO-2	Shards gradients and optimizer state across data-parallel ranks.
ZeRO-3	Shards trainable parameters as well, adding gather and prefetch windows around computation.
LoRA	Changes trainable tensors and optimizer state through adapter rank and target modules.

Among feasible results, the reported headroom gives the user a practical margin against allocator variation, dataloader details, and small differences between the synthetic estimate and the real training script. The search output therefore contains both the selected candidate and the alternatives, allowing a user to choose a slower option with more margin when the cluster allocation is scarce.

4.1 Slowdown Model

Slowdown estimation is separate from memory estimation. The web recommendation path uses a small heuristic prior:

$$R_{\text{slow}} = (1 + s_{\text{dist}})(1 + s_{\text{ckpt}})(1 + s_{\text{tp}})(1 + s_{\text{sp}}) - 1.$$

The distributed-mode priors are 0% for single-GPU and pure tensor parallelism, 3% for DDP, 8% for ZeRO-2, and 18% for ZeRO-3. Checkpointing contributes 30% because recomputation increases backward work. Tensor parallelism contributes $0.08 \log_2(T_P)$, and sequence parallelism contributes 4% when enabled.

The slowdown score is simple because estimating training time is a hard problem outside the scope of this project. It expresses the expected slowdown relative to the least-sharded feasible candidate. For example, checkpointing receives a large penalty because it trades memory for extra backward compute. ZeRO-3 receives a larger distributed penalty than ZeRO-2 because parameter gathering introduces additional synchronization and workspace.

4.2 Training Export

Search results can be exported into a TRL-compatible YAML strategy. The export contains the model reference, tuning mode, optimizer and precision choices, sequence length, checkpointing, attention backend, distributed strategy, and LoRA fields when applicable. The training runner consumes the first exported candidate by default. In the intended workflow, the user searches over model, context length, resource pool, and distributed strategy, exports the best candidate, and launches training from that file:

```
simplesft search <HF Model> ... --export-file ./trl_strategy.yaml
simplesft train --config ./trl_strategy.yaml --dataset <HF Dataset>
```

The command-line interface and web server return the same core objects: peak memory, feasibility, phase records, component breakdowns, assumptions, and diagnostic metadata.

5 Measurement and Evaluation Method

As a ground-truth target for estimator development, we measured the memory usage of various parallel training strategies. SimpleSFT runs real CUDA microsteps and records memory use around the training phases that the analytical model predicts. Measurements are collected once, saved with their configurations, and used during analytical model calibration.

DDP runs use `torchrun`. Each rank measures its own local microstep, serializes a `MemoryResult`, and rank zero stores a conservative max-across-ranks aggregate. ZeRO-2 and ZeRO-3 use a DeepSpeed launcher with the same stage, bucket, prefetch, overlap, reduce-scatter, all-gather, and precision settings used by the estimator. Single-rank ZeRO measurement is rejected because sharding requires more than one rank.

The measured breakdown records live parameter bytes, gradient bytes, optimizer-state bytes, retained activation bytes, runtime reserve, and transient remainder. Synthetic text batches keep the sequence length and label shape controlled across runs.

6 Quantitative Accuracy

The current saved measurement corpus covers 324 rows and skips 6 rows that still fail for artifact or support reasons. The global error is small enough for practical feasibility screening while leaving room for improvement in runtime lifetime modeling.

Global MAE = 1.332 GiB
Mean Absolute Relative Error = 9.1%
Median Absolute Relative Error = 6.1%
Retained-Forward Proxy MAE = 1.101 GiB

The most useful interpretation of these numbers is operational. An estimate with several GiB of headroom is usually enough to decide whether a launch configuration is plausible. Tight fits still need measured confirmation, especially when a strategy relies on ZeRO, checkpointing, or long-context attention kernels.

7 Limitations and Future Work

Several parts of the estimator still depend on calibrated runtime assumptions. Runtime support and optimizer-step carryover are fudge terms that account for memory overhead outside the analytical model’s current coverage. Additionally, the slowdown score used for recommendations should become a measured timing model.

The next technical step is to connect timing and memory in the same artifact. The memory estimator already ranks feasibility well enough for pre-launch screening. A measured timing model would make slowdown estimates less heuristic, especially for ZeRO-3, checkpointing, and tensor-parallel jobs where communication and recomputation dominate user-visible cost.

8 Related Work

SimpleSFT follows prior work on memory-aware fine-tuning and distributed training while targeting a narrower workflow: fast SFT strategy selection on research LLMs before launch. LLMem estimates memory for distributed fine-tuning and recommends methods for multi-GPU LLM training [3], but it lacks modern attention backends and support for LoRA. Empirical work on distributed Transformer training studies how model architecture and distributed strategy interact with memory and performance [4]. Broader resource-management work compares analytical, CPU-side, and learning-based estimators for GPU memory and utilization [7]. SimpleSFT uses the same general motivation, then adds project-specific measurement artifacts and SFT-oriented strategy export.

Work	Relationship to SimpleSFT
LLMem [3]	Analytical GPU-memory estimation for LLM fine-tuning and method recommendation across distributed training settings. It was calibrated using older models and different data regimes.
Transformer training study [4]	Empirical view of memory and performance behavior under distributed training choices.
Training-aware resource management [7]	Broader resource-management framing for memory and utilization prediction.

This work builds directly on prior work in training optimization. LoRA reduces trainable state by injecting low-rank adapter matrices into frozen base models [2]. ZeRO shards redundant training state across data parallel ranks [5]. FlashAttention changes the memory shape of attention through IO-aware tiling that avoids the full quadratic score matrix [1]. These methods are common in current LLM fine-tuning practice, including Qwen-family research workflows [6].

9 Conclusion

SimpleSFT gives researchers a practical way to plan SFT jobs before reserving scarce GPU time. The estimator connects inspected model structure with distributed training choices, LoRA, checkpointing, FlashAttention, optimizer state, and CUDA runtime support. The result is a phase-level estimate that can guide resource selection and clarify the expected source of memory pressure.

The current saved measurement corpus shows a global mean absolute error of 1.332 GiB and a median relative error of 6.1 percent. Those numbers are strong for pre-launch screening, especially when users keep several GiB of memory headroom.

References

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A Web Interface Screenshot

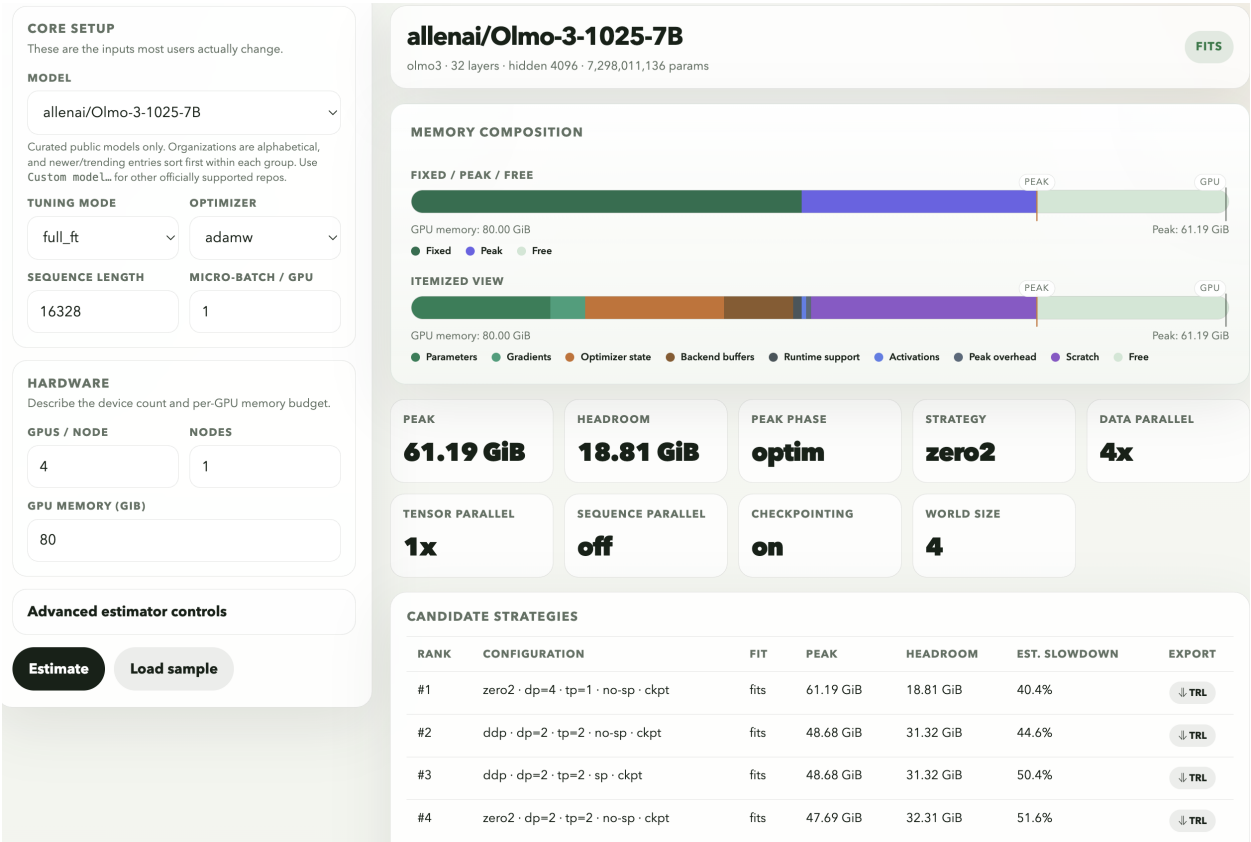


Figure 1: SimpleSFT web interface showing an example estimate and ranked candidate strategies.